

LESSON 7-5 • SESSION 2 • PRACTICE • FORM A

Reflecting Points Across Axes

Learning target – I can reflect a point across the x-axis or the y-axis and name the new coordinates. · 6.NS.C.6b

REMEMBER

- A **reflection** flips a point like a mirror image. Only **one** coordinate changes sign.
- Across the **x-axis**, y changes sign: (4, 1) becomes (4, -1). Across the **y-axis**, x changes sign: (4, 1) becomes (-4, 1).
- A point and its image are the same distance from the mirror: (-5, 2) and (5, 2) are $5 + 5 = 10$ units apart.

WORKED EXAMPLE

Reflect (7, -2) across the x-axis, then across the y-axis.

Step 1 Across the x-axis: keep $x = 7$, change -2 to 2 .
Image: (7, 2).**Step 2** Across the y-axis: keep $y = 2$, change 7 to -7 . Image: (-7, 2).**Answer (7, 2) and (-7, 2)****WORDS** reflection (*reflexión*) · **x-axis** (*eje x*) · **y-axis** (*eje y*)

PRACTICE

show your work · label units

1. Write the image after each reflection.

a. (5, 3) across the x-axis _____

b. (-2, 8) across the y-axis _____

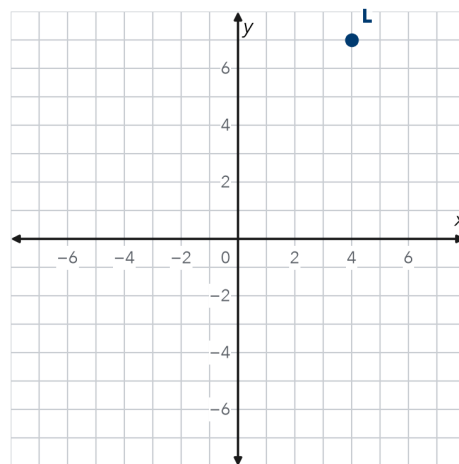
c. (7, -1) across the x-axis _____

d. (-9, -4) across the y-axis _____

2. **Mirror Room.** A floor plan puts lamp L at (4, 7). One mirror lies along each axis.**Hint** Plot each image to check your answer.

a. Across the x-axis: (____, ____) Quadrant ____

b. Across the y-axis: (____, ____) Quadrant ____

3. **Two Kites.** Kite A flies over (-6, 3). Kite B is its reflection across the y-axis. Where is Kite B? How far apart are the kites?**Hint** 6 units to the mirror, then 6 more.**Kite B, distance** _____4. **Stage Lights.** A light hangs over (-8, 12). A second light is its reflection across the x-axis. Where is it? How far apart are the lights?**Light 2, distance** _____5. **Which Mirror?** Point P is at (2, -7) and point Q is at (2, 7). Across which axis are they reflections? How do the coordinates show it?**Hint** Which coordinate changed sign?

6. **Ben's Image.** Ben says reflecting $(-3, 6)$ across the y-axis gives $(-3, -6)$. What did he mix up? Write the correct image.

Hint Across the y-axis, y stays the same.

Show your work.

Image _____

7. **Double Flip.** A marker at $(5, -8)$ reflects across the x-axis. Then that image reflects across the y-axis.
- First image: _____
 - Final image: _____
 - Quadrant of the final image: _____
8. **Design Symmetry.** A designer puts a corner at $A(3, 4)$ and wants matching corners in the other three quadrants.
- Reflect A across the y-axis. _____
 - Reflect A across the x-axis. _____
 - Name the fourth corner. How did you find it? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Lucia's Work.** To reflect $(-1, 8)$ across the x-axis, Lucia flipped both signs and wrote $(1, -8)$. Explain her mistake. Then write the correct image.

TEST PRACTICE

choose the best answer

10. Reflecting $(4, -6)$ across the x-axis gives –
- A** $(-4, -6)$ **B** $(4, 6)$ **C** $(-4, 6)$ **D** $(6, -4)$
11. Reflecting across the y-axis changes the sign of –
- A** the x-coordinate **B** the y-coordinate
C both coordinates **D** neither coordinate

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

(4, -7) E	(-3, -4) S	x-axis E	y-axis Z	(3, 6) C	(6, 3) F	(-3, -6) Q	24 L	(-5, 8) T	(5, -3) R
#1a	#2a	#3	#4	#5	#6	#7b	#8c		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____